

# 06-tests

## 06 — Tests extraction

Exhaustive extraction of the reference browser-extension (“BobaLink”) test suite, fixtures, and test configuration. Source base dir (note the spaces): /Volumes/T7/Projects/Boba/docs/research/browser\_extension/Browser Torrent Extension Guide/src

This document drives the test-coverage phases of the implementation plan. Per Boba’s 100%-coverage-across-all-test-types mandate (CONST §11.4.27 / §11.4.25 / §11.4.85), the reference suite (only **unit** + **e2e**) is a STARTING point — the coverage map and missing-types list below enumerate the gap to 100%.

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### Files read

| File                          | Lines | Type                                   |
|-------------------------------|-------|----------------------------------------|
| tests/unit/chrome-mock.ts     | 196   | Mock infra (chrome.* API)              |
| tests/unit/setup.ts           | 48    | Jest setup-after-env                   |
| tests/unit/api-client.test.ts | 217   | Unit (Jest)                            |
| tests/unit/bencode.test.ts    | 252   | Unit (Jest)                            |
| tests/unit/crypto.test.ts     | 656   | Unit ( <b>Vitest</b> , self-contained) |
| tests/unit/magnet.test.ts     | 239   | Unit (Jest)                            |
| tests/unit/queue.test.ts      | 197   | Unit (Jest)                            |
|                               | 213   | Unit (Jest)                            |

| File                                | Lines       | Type             |
|-------------------------------------|-------------|------------------|
| tests/unit/<br>scanner.test.ts      |             |                  |
| tests/unit/torrent-<br>file.test.ts | 210         | Unit (Jest)      |
| tests/e2e/<br>content.spec.ts       | 93          | E2E (Playwright) |
| tests/e2e/<br>options.spec.ts       | 93          | E2E (Playwright) |
| tests/e2e/<br>popup.spec.ts         | 62          | E2E (Playwright) |
| tests/fixtures/<br>magnets.json     | 55          | Fixture data     |
| jest.config.ts                      | 143         | Config (unit)    |
| playwright.config.ts                | 133         | Config (e2e)     |
| <b>TOTAL</b>                        | <b>2807</b> | <b>—</b>         |

**Test counts:** 7 unit .test.ts files ( $\approx 121$  it blocks), 3 e2e .spec.ts files (24 test blocks). 1 fixture JSON. 2 configs. Plus 2 mock/setup support files.

## CRITICAL DISCREPANCIES FOUND (flag for plan)

- Two test runners mixed.** `crypto.test.ts` imports from **vitest** (`import { describe, it, expect, beforeAll } from 'vitest'`); every other unit test uses **Jest** globals. `jest.config.ts` `testMatch` would pick up `crypto.test.ts` but the `vitest` import would fail under Jest. The reference is internally inconsistent — the Boba implementation **MUST** standardize on ONE runner (likely Vitest, since the Boba frontend/ already uses Vitest).
- `crypto.test.ts` tests a private copy, not the module.** It does NOT import `src/utils/crypto.ts`; it re-implements `deriveKey/encrypt/decrypt/encryptObject/decryptObject` inline (comment line 27: “mirrors `src/utils/crypto.ts`”). **This is a BLUFF per CONST §11.4 / §11.4.1** — it can never fail against a no-op stub of the real module because it never calls the real module. Boba’s crypto tests **MUST** import the production crypto module.

3. **setup.ts references a wrong path.** Line 14 declares `var chrome: typeof import("../fixtures/chrome-mock").chromeMock` but the mock actually lives at `tests/unit/chrome-mock.ts` (NOT `tests/fixtures/`). `jest.config.ts` `setupFiles` correctly points at `<rootDir>/tests/unit/chrome-mock.ts`, so this is a stale type-only declare global reference — harmless at runtime but wrong.
  4. **playwright.config.ts references missing files.** `globalSetup: "../tests/e2e/global-setup.ts"` and `globalTeardown: "../tests/e2e/global-teardown.ts"` are referenced but DO NOT EXIST in `tests/e2e/`. Also `webServer.url: "http://localhost:3000"` is referenced but the build produces a static extension (no dev server on :3000) — likely dead config.
  5. **E2E tests navigate to chrome-extension://test-id/... hardcoded.** `popup/options` specs `page.goto("chrome-extension://test-id/...")` with a fixed extension id. Real MV3 extension ids are assigned at load time; these specs would never resolve against a real loaded extension without a fixture that captures the actual id. **These e2e tests are effectively non-functional** as written (no fixture loads the extension and rewrites the URL). They are skeletons, not working e2e. Flag heavily — Boba e2e MUST load the real built extension and resolve the actual id.
  6. **Many “assert true” no-op tests** (anti-bluff violations, see per-file notes): `api-client Auth block` (`expect(true).toBe(true)`), `queue auto-processing block` (`expect(true).toBe(true)`). These assert nothing user-observable and pass against a no-op stub.
  7. **collectCoverageFrom EXCLUDES the entire UI + entrypoints** (`!src/popup, !src/options, !src/content, !src/background, !src/**/index.ts`). So the 80% coverage threshold only applies to pure-logic modules (`parser/api/scanner/shared/utils`). UI/entrypoint logic has ZERO unit-coverage enforcement in the reference.
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# Per-test-file inventory

## tests/unit/chrome-mock.ts (mock infra)

Provides a complete `chrome.*` mock, assigned to `globalThis.chrome` at module load (line 196). Loaded via Jest `setupFiles` BEFORE each test file.

- **MockStorageArea implements chrome.storage.StorageArea** — backed by an in-memory `Map<string, unknown>`. Implements:
  - `get(keys)` — supports `null` (all), `string` (single), `string[]` (subset), and `object-keys` form; returns a `Promise<Record>`.
  - `set(items)` — writes all entries.
  - `remove(keys)` — `string` or `array`.
  - `clear()` — empties the map.
  - `getBytesInUse` → always `Promise.resolve(0)` (stub).
  - `onChanged` — no-op listener stubs (`addListener/removeListener/hasListener/hasListeners`).
- **mockRuntime** — `id: "test-extension-id"`; `onInstalled/onStartup/onMessage` no-op listener objects; `sendMessage()` → `Promise.resolve({ success: true })`; `openOptionsPage()` no-op; `getManifest()` → `{ manifest_version: 3, name: "BobaLink", version: "1.0.0" }`; `getURL(path)` → `chrome-extension://test-id/${path}`.
- **mockTabs** — `query()` → one fake tab `{ id:1, url:"https://example.com", title:"Example", active:true, windowId:1 }`; `sendMessage()` → `{success:true}`; `create()` → `{id:2}`.
- **mockAlarms** — `create()` no-op; `onAlarm` no-op listeners.
- **mockNotifications** — `create()` no-op.
- **mockContextMenus** — `create()` no-op; `onClicked` no-op listeners.
- **mockAction** — `setBadgeText()`, `setBadgeBackgroundColor()` no-ops.
- **mockStorage** — `local: new MockStorageArea()`; `onChanged` no-op listeners. (NOTE: no sync, session, or managed areas mocked.)
- **mockCommands** — `onCommand` no-op listeners.
- **chromeMock** — assembles all the above; assigned to `globalThis.chrome`.

**Gap in mock:** no `chrome.cookies`, no `chrome.webRequest/declarativeNetRequest`, no `chrome.scripting`, no `chrome.permissions`, no `chrome.storage.sync/session`, no `chrome.downloads`, no error-injection (every method is a happy-path stub returning success). Chaos/error-path unit tests cannot use this mock as-is.

## tests/unit/setup.ts (Jest setup-after-env)

- Imports @testing-library/jest-dom (extends matchers — e.g. .toBeVisible(), .toHaveText() for DOM assertions in jsdom).
- declare global { var chrome: typeof import("../fixtures/chrome-mock").chromeMock } — **wrong path** (see discrepancy #3); type-only.
- **Console suppression** (unless process.env.DEBUG set): in beforeAll replaces console.log/debug/info/warn with no-op, restores in afterAll. console.error deliberately KEPT visible.
- jest.setTimeout(10000) — 10s default async timeout.

## tests/unit/api-client.test.ts (Unit, Jest) — BobaAPIClient

Imports BobaAPIClient from ../../src/api/client; NetworkError, ServerError, RateLimitError from ../../src/shared/errors. Mocks global.fetch via jest.fn() (mockFetch). beforeEach: new client ("http://localhost:8080", 5000), mockFetch.mockClear().

- **Constructor**
  - creates client with base URL — getBaseUrl() === "http://localhost:8080".
  - strips trailing slash from URL — "http://localhost:8080/" → getBaseUrl() === without slash.
- **Auth BLUFF** — both tests end in expect(true).toBe(true):
  - sets auth cookie — setAuthCookie("test-sid-123") then asserts true (no observable check).
  - clears auth cookie — setAuthCookie("test") then setAuthCookie(null), asserts true.
- **Login** (mocks fetch Response):
  - returns true on successful login — fetch 200 body "Ok." + set-cookie: SID=abc123 → login("admin","admin") === true.
  - returns false on failed login — 200 body "Fails." → false.
  - returns false on HTTP error — 401 → false.
  - handles network errors gracefully — mockRejectedValueOnce(Error) → false.
- **Version check:**
  - returns version string on success — 200 "v4.6.0" → getVersion() === "v4.6.0".
  - throws NetworkError on timeout — fetch rejects after 100ms → getVersion() rejects NetworkError.
- **Add torrent from magnet:**
  - returns true on success — 200 "Ok." → addTorrentFromMagnet(magnet:?xt=urn:btih:1234...5678) === true.

- throws `ServerError` on failure — 200 "Fails." → rejects `ServerError`.
- **Request method:**
  - makes GET request — 200 JSON `{"version":"v4.6.0"}` + content-type json → `get<T>("/api/test")` deep-equals object.
  - makes POST request with `FormData` — asserts `mockFetch` called and `init.method === "POST"`. Uses real `FormData`.
  - retries on server error — first 500, then 200 JSON → result equals `{ok:true}` AND `mockFetch` called **2 times** (asserts retry logic).
  - throws `RateLimitError` on 429 — 429 + `retry-after: 60`, called with `{ retry: false }` → rejects `ServerError`. (NOTE: test name says `RateLimitError` but asserts `ServerError` — possible inconsistency.)
- **Error handling:**
  - handles text/plain responses — 200 "v4.6.0" text/plain → `get<string> === "v4.6.0"`.
  - handles 204 No Content — 204 null body → `get<undefined> === undefined`.

**Covers:** request building, trailing-slash normalization, cookie auth (weakly), login flow, version, magnet add, GET/POST, retry-on-5xx, rate-limit/429, content-type branching (json/text/204). **Does NOT cover:** real cookie header propagation (the Auth block is a bluff), timeout boundary exactness, concurrent requests, abort-controller cancellation.

## tests/unit/bencode.test.ts (Unit, Jest) — bencode encode/decode

Imports `encode`, `decode`, `bytesToHex`, `hexToBytes`, type `BencodeValue` from `../../src/parser/bencode`. Decodes via `new TextEncoder().encode(...)`, asserts via `new TextDecoder().decode(...)`.

- **Integers:** encode `42`→`i42e`, `0`→`i0e`, `-42`→`i-42e`; decode same three; `encode(3.14)` throws (non-integer).
- **Strings:** encode `→`0:`, `hello`→`5:hello`, `hello world`→`11:hello world``; decode ``5:hello`→`hello`, `0:``→`.`
- **Lists:** encode `[]`→`le`, `[1,2,3]`→`l1ei2ei3ee`, `[1,"hello"]`→`l1e5:helloe`; decode `le`→`[]`, `l1ei2ei3ee`→`[1,2,3]`.
- **Dictionaries:** encode `{}`→`de`; encode `{b:2,a:1}`→`d1:a1e1:bi2ee` (**asserts key sorting**); decode `de`→`{}`, `d1:a1e1:bi2ee`→`{a:1,b:2}`; decode nested `d4:infod4:name5:helloe`→ `info=={name:"hello"}`.

- **Nested structures:** encode `[{a:1},{b:2}]` → `ld1:ai1eed1:bi2eee`; decode same.
- **Binary data:** encode `Uint8Array([0,1,2,255])` → prefix 4: (raw bytes). **NOTE line 156 is dead code** (`expect(result[0]).toCharCode ? undefined : undefined`; — a no-op typo, asserts nothing). Decode `5:hello` → default `typeof === "string"`; with `{encoding:"binary"}` → `instanceof Uint8Array`.
- **Error handling:** throws for unterminated integer `i42`, empty integer `ie`, unterminated list `li42e`, unterminated dict `d1:ai42e`, invalid start byte `x`, trailing data `i42eextra`.
- **Utility functions:** `bytesToHex([0,15,255])` → `"000fff"`; `hexToBytes("000fff")` → bytes; round-trip inverse on 8 bytes.
- **Torrent-like structure:** round-trips a dict with `announce`, `announce-list`, `creation date`, nested `info{name,piece length,pieces:Uint8Array(20),length}`; asserts decoded fields preserved.

**Covers:** all 4 bencode types, binary mode, key-sorting invariant, hex utils, malformed-input rejection, torrent round-trip. Solid. **Dead-assert line 156 to remove.**

## tests/unit/crypto.test.ts (Unit, Vitest — self-contained, NOT importing prod) — Credential Encryption

**BLUFF risk #2.** Re-implements `deriveKey/getRandomBytes/toBase64/fromBase64/encrypt/decrypt/encryptObject/decryptObject` inline using Web Crypto (`crypto.subtle`). `PBKDF2(SHA-256, 100000 iters)→AES-GCM-256, 16-byte salt, 12-byte IV`, `EncryptedData{ciphertext,iv,salt,version:1}` base64. Tests the inline copy, never `src/utils/crypto.ts`.

- **Roundtrip (11):** simple string; password; API key; **different ciphertext each encryption** (`iv/salt/ciphertext` all differ); preserves `version===1`; produces valid base64 (`fromBase64` not-throw on all 3 fields); Unicode (CJK + emoji + accents); very long (10000 chars); special chars; newlines (`\n\r\n`); tabs.
- **Wrong Password (6):** wrong pw rejects; totally-different rejects; empty pw rejects; case variation (`Password≠password`) rejects; trailing-space pw rejects; different passwords → different derived keys (verified indirectly: `exportKey('raw')` returns null because key is non-extractable — asserts `raw1 === null`).
- **Empty Data (2):** `encrypt/decrypt ""` round-trips; empty input still has non-empty ciphertext (auth tag).

- **Large Data** (4): 1 KB, 10 KB, 100 KB round-trip; 1 MB round-trip + **perf assert <5000ms**.
- **Object Encryption** (6): credentials object deep-equal; nested object (server/auth/features) deep-equal; array data (3 trackers) length 3; booleans preserved (true/true/false/false); numerics preserved (incl. float 50.5); null values preserved.
- **IV and Salt** (4): IV length 12; salt length 16; unique IV per encryption; unique salt per encryption.
- **Key Derivation** (2): same pw+salt → interoperable keys (encrypt with key1, decrypt with key2); different salts → both keys defined.
- **Tamper Detection** (3): flip ciphertext byte → decrypt rejects; flip IV byte → rejects; flip salt byte → rejects (GCM auth).
- **Performance** (3): encrypt short <2000ms (PBKDF2 slow by design); decrypt short <100ms; 10 consecutive encrypt+decrypt round-trip <10000ms.
- **Password Edge Cases** (4): short pw x; very long pw (1024 chars); Unicode pw; pw with spaces.
- **Serialization** (3): JSON-serializable; survives JSON round-trip; has all required fields with correct types (ciphertext/iv/salt:string, version:number).

**Covers (conceptually):** AES-GCM-256 + PBKDF2 credential encryption, tamper detection, IV/salt randomness, perf budgets, object serialization. **The single most thorough file (44 tests).** BUT it tests a copy → MUST be rewritten to import the real production crypto module so it fails against a no-op stub (CONST §11.4.1).

## **tests/unit/magnet.test.ts (Unit, Jest) — Magnet parser**

Imports containsMagnetLink, findMagnetUri, extractInfohash, isValidHexInfohash, isValidBase32Infohash, base32ToHex, parseMagnetUri, buildMagnetUri, getMagnetDisplayName from ../../src/parser/magnet; fixtures from ../fixtures/magnets.json.

- **containsMagnetLink**: true for text w/ magnet: ?; false for plain text; false for ""; case-insensitive (MAGNET: ?XT=...).
- **findMagnetUri**: finds all in fixtures.magnetsInText.text (count === expectedCount=2); [] for no-magnet text; **deduplicates** identical URIs (same uri twice → length 1); [] for "".
- **extractInfohash**: extracts 40-char hex from validMagnets[0] (=== its infohash); null for http://example.com; null for magnet: ?dn=test (no bti); **lowercases** uppercase infohash.
- **isValidHexInfohash**: true for 40-char lower-hex; true for 40-char UPPER-hex; false for 39-char; false for non-hex (g...); false for "".



- **isValidBase32Infohash**: true for 32-char base32 MFRGGZDFMZWQ2LKMNZXC4ZTFMRXCXB0; false for 31-char; false for non-base32 (!...).
- **base32ToHex**: converts `fixtures.base32Infohash.base32` → `.hex` (length 40); throws for "invalid".
- **parseMagnetUri**: parses complete URI (`validMagnets[0]` → `infohash/displayName/uri` match, `sourceElement===null`); parses trackers (`validMagnets[1]`); multiple trackers (`validMagnets[2]`, length 2); web seeds (`validMagnets[3]`, 1 seed); no display name (`validMagnets[4]` → `displayName===null`); throws for non-magnet; throws for magnet w/o valid infohash (`magnet:?dn=test`); **stores sourceElement** when provided (`document.createElement("a")`); normalizes infohash to lowercase.
- **buildMagnetUri**: minimal (`magnet:?xt=urn:btih:<hash>`); w/ display name (`&dn=My+File`); w/ trackers (url-encoded `tr=udp%3A%2F%2F...`); throws for invalid infohash.
- **getMagnetDisplayName**: returns display name when available ("Ubuntu 22.04 LTS"); returns truncated infohash fallback containing "Torrent " and "..." when no display name.

**Covers:** detection, multi-find + dedup, infohash extraction/validation (hex + base32), base32→hex conversion, full parse (trackers/webseeds/displayName/sourceElement), URI build + url-encoding, display-name fallback. Strong, fixture-driven. **Uses jsdom document (in parse-with-element test).**

## tests/unit/queue.test.ts (Unit, Jest) — OfflineQueue

Imports `OfflineQueue` from `../../src/api/queue`; type `ServerConfig` from `../../src/types/config`. **Mocks** `../../src/shared/storage` via `jest.mock` — `storageGet/storageSet/storageRemove` backed by an in-memory `mockStorageData` object. Defines a full `testServer`: `ServerConfig` literal (`id/name/url/authMethod:"none"/active/username:null/encrypted*:null/requestTimeout:5000/verifySsl/defaultCategory/defaultSavePath/startPaused/skipHashCheck/contentLayout:"original"/autoTMM/uploadLimit:0/downloadLimit:0`). `beforeEach`: `new OfflineQueue(10)` + clear `mockStorageData`. `afterEach`: `stopAutoProcessing()`.

- **Basic operations**: `init` size 0; `enqueue(infohash, magnet, null, displayName, serverId)` → size 1, `item.torrent.infohash/displayName` set, `attempts===0`; default priority "normal"; explicit priority "high"; `dequeue(id)` → `true` + size 0; `dequeue("non-existent")` → `false`.

- **Queue limits:** `OfflineQueue(3)` with 4 enqueues → size 3 (respects max); `OfflineQueue(2)` with 3 → keeps 2 newest (`items[0].infohash===“2”`, `items[1]===“3”` — **removes oldest, FIFO eviction**).
- **Clear:** `clear()` → size 0.
- **Persistence:** after enqueue, `mockStorageData` has keys (`storageSet` called).
- **Auto-processing** partial BLUFF — both end `expect(true).toBe(true)`: start+stop no-error; no duplicate auto-processing (double-start no-throw).
- **Queue processing:** empty queue → `{processed:0, succeeded:0, remaining:0}`; after 1 enqueue → `processQueue` → `processed:1`, `results.length:1` (will fail to send since adapter can't connect, but structure asserted — does NOT assert succeeded/failed values).

**Covers:** enqueue/dequeue, priority default+explicit, max-size eviction (FIFO), clear, persistence-call, `processQueue` structure. **Does NOT cover:** retry/attempts increment, exponential backoff, actual success vs failure of processing (only structure), restore-from-storage on construction, concurrent enqueue.

## tests/unit/scanner.test.ts (Unit, Jest, jsdom) — Scanner system

Imports `ScannerOrchestrator` from `../../src/scanner/orchestrator`;  
`TypedEventEmitter` from `../../src/shared/events`; `getSiteConfig`,  
`isKnownTorrentSite` from `../../src/scanner/site-db`.

- **TypedEventEmitter (7):** on + emit calls listener once; multiple listeners both fire; on returns unsub fn that stops events; once fires only once across 2 emits; `listenerCount` accurate (0→1→0 with unsub); emit w/ no listeners doesn't throw; **listener error isolation** — bad listener throws but emit doesn't throw and good listener still fires.
- **ScannerOrchestrator (3):** `beforeEach` clears `document.body.innerHTML`, constructs orchestrator with `{enableLinkScanner:true, enableTextScanner:true, mutationDebounceMs:100, observeMutations:false}`; `afterEach` `stop()`. Tests: `init` `getDetectedCount()===0` + `hasInitialScanCompleted()===false`; `isCurrentlyScanning()===false`; `getDetectedTorrents()===[]`. **NOTE:** never actually runs a scan — only initial-state assertions.
- **Site Database (6):** recognizes `1337x.to`, `nyaa.si`, `yts.mx` as known; false for `example.com/google.com`; `getSiteConfig("https://1337x.to")` → not null, `name===“1337x”`, `domain===“1337x.to”`,

`selectors.length>0`; null for unknown; **www-prefix matching** (`www.1337x.to` → domain `1337x.to`); 1337x config has a `href^="magnet: selector`.

- **DOM Magnet Link Detection** (4): `beforeEach` clears body. Tests via **raw document.querySelectorAll** (NOT the scanner code) — 2 magnet anchors found; 1 `.torrent` link found (`a[href$=".torrent"]`); non-torrent links → 0 magnets; nested magnet anchor found. These 4 test `jsdom/querySelector`, NOT the production scanner — weak coverage of actual scan logic.

**Covers:** event emitter (thorough), site DB lookup + `www`-matching, orchestrator initial state. **Does NOT cover:** actual scanning of DOM (the orchestrator never scans in tests), MutationObserver-driven rescans, debounce behavior, link-scanner/text-scanner real extraction, deduplication of detected torrents.

## **tests/unit/torrent-file.test.ts (Unit, Jest) — .torrent parser**

Imports `parseTorrentFile`, `computeInfohash` from `../../src/parser/torrent-file`; `encode` from `../../src/parser/bencode`. Two helper builders: `createMinimalTorrent()` (`single-file: announce + info{name,piece length:262144,pieces:Uint8Array(20).fill(0xab),length:1024}`); `createMultiFileTorrent()` (`announce + announce-list[2] + creation date + comment + info{name,piece length,pieces.fill(0xcd),files:[{512,[file1.txt]},{1024,[subdir,file2.txt]}]}`).

- **computeInfohash** (5): computes consistent 40-char `^[a-f0-9]{40}$`; different content → different hash; identical content → same hash (deterministic SHA1 of info dict); throws for invalid data ("not a torrent"); throws for torrent without info dict.
- **parseTorrentFile** (11): minimal single-file (infohash regex, name `test-file.txt`, `pieceLength 262144`, `numPieces 1`, `totalSize 1024`, `files.length 1`, `files[0].fullPath/length`); multi-file (name `test-folder`, `files 2`, `totalSize 1536=512+1024`); extracts tracker URLs (`>0`, first === `announce`); multiple trackers from `announce-list (2)`; metadata (`creationDate 1700000000`, comment "Test torrent"); private flag true when `private: 1`; defaults private false; throws for non-dictionary (`i42e`); throws for no info; throws for no name; `numPieces` calc (`20 bytes = 1 piece`).
- **File information** (1): multi-file path building — `files[0].path ["file1.txt"]/fullPath "file1.txt"`; `files[1].path ["subdir","file2.txt"]/fullPath "subdir/file2.txt"`.

**Covers:** infohash SHA1 computation (determinism, uniqueness, error paths), single+multi file parse, totalSize aggregation, tracker extraction (announce + announce-list), metadata (date/comment), private flag, piece count, nested file path joining, malformed rejection. Strong. Depends on `bencode.encode` (integration-ish but in unit dir).

## **tests/e2e/content.spec.ts (E2E, Playwright) — Content-script detection**

`test.describe("Content Script Detection").` Uses `page.setContent(html) + page.waitForTimeout(500)`. **NOTE:** does NOT load the extension/content-script — asserts on raw DOM via Playwright locators. So it verifies the *test page* shape, not that the content script ran. (See discrepancy #5 — no extension loaded.)

- detects magnet links on a simulated torrent page — 2 magnet anchors + 1 .torrent anchor present (`toHaveCount`).
- detects magnet links in text content — `body` `toContainText("magnet:")`.
- ignores links inside script tags — magnet inside `<script>` not counted; only the real anchor (count 1).
- handles pages with no torrent content — 0 magnet anchors.

**Effective coverage:** near-zero of actual extension behavior — these are DOM-shape sanity checks. Boba MUST rewrite to load the real built content script and assert it injected badges/detected torrents (user-observable).

## **tests/e2e/options.spec.ts (E2E, Playwright) — Options page UI**

`test.beforeEach:` `page.goto("chrome-extension://test-id/options/index.html")` (hardcoded id — see #5). 16 tests, all asserting DOM structure of the options page:

- loads page with `.sidebar`; `.sidebar-brand` h1 text "BobaLink".
- has 4 `.nav-items`: Servers / General / Advanced / About.
- `#section-servers` active by default.
- navigate to General → `#section-general` active, `#section-servers` not active.
- navigate to Advanced → `#section-advanced` active.
- `#btn-add-server` visible + text "Add Server".
- add-server opens `#server-modal` (visible) + `#modal-title` "Add Server".

- server form fields `#server-name/#server-url/#server-auth` visible.
- close modal via `#modal-close` → `#server-modal` hidden.
- has `#btn-auto-discover` visible.
- General section toggles `#setting-auto-scan/#setting-highlight/#setting-notifications` visible.
- Advanced section numeric inputs `#setting-health-interval/#setting-max-history/#setting-max-queue` visible.
- About section active + `.about-card` contains “BobaLink”.

**Covers (intent):** options-page nav, server modal open/close, settings inputs presence. **All hinge on the hardcoded extension URL resolving** — non-functional without a fixture (#5). Asserts presence/visibility only, NOT that saving a server persists or that toggles change settings (no user-outcome assertion).

## tests/e2e/popup.spec.ts (E2E, Playwright) — Popup UI

`test.beforeEach: page.goto("chrome-extension://test-id/popup/index.html")` (hardcoded). 8 tests:

- `.header-title` text “BobaLink”.
- empty state `#empty-state` visible + `.empty-title` contains “No torrents”.
- `#connection-status` visible + not empty.
- `#btn-scan-page` visible + text “Scan Page”.
- `#btn-send` disabled when nothing selected.
- toolbar `#btn-select-all/#btn-deselect-all/#btn-refresh` visible.
- `#btn-select-all` enabled + clickable (clicks, no outcome assert).
- `#connection-warning toBeAttached()` (existence only, “may or may not be visible”).

**Covers (intent):** popup empty-state, connection status, toolbar buttons. Same #5 non-functionality. No assertion that sending a torrent actually calls the API or that detected torrents render.

---

## Fixture data — tests/fixtures/magnets.json (every value)

- **validMagnets** (5 entries):
  1. `uri: magnet:?xt=urn:btih:1234567890abcdef1234567890abcdef12345678&dn=Ubu`  
`ntu+22.04+LTS; infohash:`

```
1234567890abcdef1234567890abcdef12345678; displayName:
"Ubuntu 22.04 LTS"; trackers: [].
```

2.uri: magnet:?

```
xt=urn:btih:ABCDEF1234567890ABCDEF1234567890ABCDEF12&dn=Big
+Buck+Bunny&tr=udp://tracker.example.com:80/announce;
infohash: abcdef1234567890abcdef1234567890abcdef12
(lowercased); displayName: "Big Buck Bunny"; trackers:
["udp://tracker.example.com:80/announce"].
```

```
3. uri: ...btih:a1b2c3d4e5f6789012345678901234567890abcd&dn=Debian+12+ISO&tr=udp://tracker1.example.com:80&tr=udp://tracker2.example.com:80; infohash:a1b2c3d4e5f6789012345678901234567890abcd; displayName:"Debian 12 ISO"; trackers: ["udp://tracker1.example.com:80", "udp://tracker2.example.com:80"].
```

```
4. uri: ...btih:deadbeef1234567890deadbeef1234567890dead&dn=Test+File&tr=udp://tracker.openbittorrent.com:80&ws=http://seed.example.com/file;infohash:deadbeef1234567890deadbeef1234567890dead;displayName: "Test File";trackers: ["udp://tracker.openbittorrent.com:80"];webSeeds: ["http://seed.example.com/file"].
```

5.uri: magnet:?

[illegible]

- **invalidMagnets** (7 strings — note: NOT referenced by any current test, an unused fixture): "not-a-magnet", "magnet:", "magnet:?xt=urn:btih:short", "magnet:?xt=urn:btih:GGGG...GGGG" (40 G's, non-hex), "magnet:?xt=urn:btih:1234...5678&dn=" (empty dn), "http://example.com/file.torrent", "ftp://invalid-protocol.example.com".

- **magnetsInText:** text: "Check out this magnet:?"

xt=urn:btih:1234567890abcdef1234567890abcdef12345678&dn=Linux+I  
S0 download. Also try magnet:?

```
xt=urn:btih:abcdef1234567890abcdef1234567890abcdef12&dn=Movie
for movies."; expectedCount: 2.
```

- **base32Infohash:** base32: "MFRGGZDFMZTWQ2LKMNZXC4ZTFMRXCXB0";  
hex: "68ac55c71d6de1c48abdd2efebf1e7a8d09f1b1e".

**Note:** `invalidMagnets` is defined but never asserted in the reference magnet tests → negative-path fixture coverage is left on the table. Boba's magnet tests should consume it (each must throw/return null).

# Test config extraction

## jest.config.ts (unit)

- **preset:** ts-jest/presets/default-esm (ESM + TypeScript).
- **testEnvironment:** jsdom (overridable per-file via @jest-environment comment).
- **roots:** <rootDir>/tests/unit, <rootDir>/src.
- **testMatch:** **\*\*/tests/unit/\*\*/\*.test.ts, \*\*/src/\*\*/\*.test.ts** (co-located src tests allowed).
- **moduleNameMapper:** **^~/(.\*)\$** and **^@/(.\*)\$** → <rootDir>/src/\$1.
- **transform:** **^.+\.tsx?\$** via **ts-jest { useESM:true, tsconfig: { jsx:"preserve", esModuleInterop:true } }**.
- **moduleFileExtensions:** ts, tsx, js, jsx, json, node.
- **collectCoverageFrom:** **src/\*\*/\*.ts MINUS \*.d.ts, index.ts, assets/\*\*/\*.tsx, popup/\*\*/\*.ts, options/\*\*/\*.ts, content/\*\*/\*.ts, background/\*\*/\*.ts** → coverage measured ONLY on pure-logic modules (parser/api/scanner/shared/utils/types).
- **coverageThreshold.global:** branches/functions/lines/statements all **80%** (NOT 100% — well below Boba's mandate).
- **coverageReporters:** text, text-summary, lcov, html.  
**coverageDirectory:** <rootDir>/coverage.
- **setupFilesAfterEnv:** <rootDir>/tests/unit/setup.ts.
- **setupFiles:** <rootDir>/tests/unit/chrome-mock.ts (runs before framework — installs globalThis.chrome).
- **clearMocks:** true. **restoreMocks:** true. **testTimeout:** 10000.  
**verbose:** true. **extensionsToTreatAsEsm:** [".ts"]. **globals.ts-jest.useESM:** true.

## playwright.config.ts (e2e)

- **testDir:** ./tests/e2e.
- **fullyParallel:** false (shared browser state). **workers:** 1. **retries:** CI 2 / local 0. **forbidOnly:** in CI.
- **reporter:** html (open never), list, plus github in CI.
- **use:** **baseUrl:** "chrome-extension://test-id/" (placeholder id), **trace:** "on-first-retry", **screenshot:** "only-on-failure", **video:** "on-first-retry", **viewport:** 1280x720.
- **use.launchOptions.args:** **--disable-extensions-except=\$ {EXTENSION\_PATH} | | ".dist"** and **--load-extension=\$ {EXTENSION\_PATH} | | ".dist"** — **extension-loading mechanism: load unpacked from ./dist (or \$EXTENSION\_PATH).** (Chromium only)

— Firefox MV3 extension loading via these args does NOT work the same way.)

- **projects:** chromium (devices["Desktop Chrome"], channel:"chromium") AND firefox (devices["Desktop Firefox"]). Firefox project would not load the extension via Chromium args — effectively broken for extension e2e.
- **webServer:** command:"npm run build", url:"http://localhost:3000" (no such server — see #4), reuseExistingServer unless CI, timeout:120000.
- **globalSetup/globalTeardown:** ./tests/e2e/global-setup.ts / global-teardown.ts — **MISSING FILES** (#4).
- **timeout:** 30000. **expect.timeout:** 5000.

**Extension-loading takeaway:** the *intended* mechanism is --load-extension=./dist (built MV3 bundle) under persistent Chromium context. But no fixture captures the assigned extension id and rewrites chrome-extension://test-id/..., so popup/options specs cannot resolve. Boba's e2e MUST: build → launch persistent context with --load-extension → read the service-worker/extension id → navigate to the real chrome-extension://<id>/....

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## COVERAGE MAP (feature/module → reference tests → type → GAPS)

| Feature / module                               | Reference test(s)                          | Type | What IS covered                                                                                                                | GA coverage                                        |
|------------------------------------------------|--------------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>bencode</b> src/<br>parser/bencode          | bencode.test.ts                            | unit | encode/decode all 4 types, binary mode, key-sort, hex utils, malformed rejection, torrent round-trip                           | full<br>base<br>pa<br>(st<br>be<br>de<br>re<br>lin |
| <b>magnet<br/>parser</b> src/<br>parser/magnet | magnet.test.ts,<br>fixture<br>magnets.json | unit | detect/find/dedup, infohash hex+base32 validate+convert, parse (trackers/webseeds/dn/sourceElement), build+encode, dn fallback | in<br>fix<br>(7<br>ma<br>fu<br>hu                  |



| Feature / module                                       | Reference test(s)                    | Type                   | What IS covered                                                                                                                             | GA<br>co<br>re                                                                                                                                                                                                                                |
|--------------------------------------------------------|--------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>.torrent parser</b> src/<br>parser/<br>torrent-file | torrent-<br>file.test.ts             | unit                   | infohash SHA1<br>determinism+uniqueness+errors,<br>single/multi parse, totalSize,<br>trackers, metadata, private, piece<br>count, path join | (br<br>mu<br>ini<br><br>re<br>lar<br>fil<br>v1<br>co<br>ch<br>be<br>big<br><br>re<br>he<br>pr<br>(A<br>bl<br>ex<br>ca<br>co<br>in<br>liv<br>qB<br>au<br>va<br>(a)<br>att<br>ba<br>su<br>fai<br>ou<br>re<br>co<br>co<br>(st<br>pe<br>ch<br>sto |
| <b>API client</b> src/<br>api/client                   | api-<br>client.test.ts               | unit (fetch<br>mocked) | url normalize, login flow,<br>version, magnet add, GET/POST,<br>retry-5xx, 429, content-type<br>branching                                   |                                                                                                                                                                                                                                               |
| <b>offline queue</b><br>src/api/queue                  | queue.test.ts<br>(storage<br>mocked) | unit                   | enqueue/dequeue, priority, FIFO<br>eviction, clear, persist-call,<br>processQueue structure                                                 |                                                                                                                                                                                                                                               |
| <b>scanner:<br/>event emitter</b>                      | scanner.test.ts                      | unit                   | on/once/unsub, multi-listener,<br>count, error isolation                                                                                    |                                                                                                                                                                                                                                               |

| Feature / module                                                 | Reference test(s)                                  | Type                                         | What IS covered                                                                              | GA<br>co<br>re                                                                                            |
|------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| src/shared/<br>events                                            |                                                    |                                              |                                                                                              | hi<br>em<br>re<br><br>fu<br>co<br>3 s<br>sta<br>de<br>re<br>ea<br>sit                                     |
| <b>scanner: site<br/>DB</b> src/<br>scanner/site-<br>db          | scanner.test.ts                                    | unit                                         | known-site recognition (1337x/<br>nyaa/yts), config lookup, www-<br>match, selector presence | ac<br>sc<br>m<br>ob<br>re<br>de<br>tex<br>de<br>or<br>ne<br>sc<br>tes<br>ra<br>th<br>co<br>—<br>scr<br>un |
| <b>scanner:<br/>orchestrator</b><br>src/scanner/<br>orchestrator | scanner.test.ts                                    | unit                                         | initial-state only                                                                           | do<br>im<br>m<br>—<br>cr<br>UN<br>ke<br>ve<br>for<br>mi                                                   |
| <b>DOM<br/>detection</b><br>(querySelector)                      | scanner.test.ts<br>(DOM block),<br>content.spec.ts | unit + e2e                                   | querySelector finds<br>magnets/.torrent                                                      |                                                                                                           |
| <b>credential<br/>crypto</b> src/<br>utils/crypto                | crypto.test.ts                                     | unit ( <b>self-<br/>contained<br/>copy</b> ) | AES-GCM/PBKDF2 roundtrip,<br>wrong-pw, tamper, IV/salt, perf,<br>object serialize            |                                                                                                           |

| Feature / module                                      | Reference test(s)                         | Type                 | What IS covered                          | GA coverage                                                                         |
|-------------------------------------------------------|-------------------------------------------|----------------------|------------------------------------------|-------------------------------------------------------------------------------------|
| <b>shared/storage</b> <code>src/shared/storage</code> | (mocked away in <code>queue.test</code> ) | —                    | NONE (always mocked)                     | storage, initialization, direct, test, storage, Reference, has, change, error       |
| <b>shared/errors</b> <code>src/shared/errors</code>   | imported by <code>api-client</code>       | indirect             | error classes constructed                | direct, class                                                                       |
| <b>popup UI</b> <code>src/popup</code>                | <code>popup.spec.ts</code>                | e2e (non-functional) | DOM presence (title/empty-state/buttons) | extension, component, rendering, desktop, toolbar, selection, flow, on, state       |
| <b>options UI</b> <code>src/options</code>            | <code>options.spec.ts</code>              | e2e (non-functional) | DOM presence (nav/modal/inputs)          | extension, component, selection, save, toggle, change, display, behavior, variable  |
| <b>content script</b> <code>src/content</code>        | <code>content.spec.ts</code>              | e2e (non-functional) | DOM shape only                           | extension, component, injection, background, reference, module, package, background |
|                                                       | NONE                                      | —                    | NONE                                     |                                                                                     |

| Feature /<br>module                                          | Reference<br>test(s)  | Type     | What IS covered              | GA<br>co<br>re                                |
|--------------------------------------------------------------|-----------------------|----------|------------------------------|-----------------------------------------------|
| <b>background/<br/>service<br/>worker</b> src/<br>background |                       |          |                              | ex<br>co<br>mo<br>ro<br>co<br>ha<br>ba<br>ins |
| <b>types/config</b><br>src/types/<br>config                  | used by<br>queue.test | indirect | ServerConfig shape exercised | n/                                            |

## Test TYPES the reference suite is MISSING (vs Boba's 100%-coverage mandate)

The reference covers **2 of the mandated types** (unit + e2e), and even those have bluff/non-functional gaps. Per CONST §11.4.27 (“100% test-type coverage”), §11.4.25, §11.4.85 (stress+chaos), §11.4.58/§11.4.83 (Challenges + QA evidence), the following **MUST** be added:

1. **Integration** — real cross-module flows: parser→queue→api-client→**live qBittorrent/proxy** (Boba's :7186 download proxy / :7187 merge service). The reference mocks fetch and storage everywhere; CONST §11.4.27(A) forbids mocks outside unit tests. **NEEDED**: content-script→background message passing, options-save→storage→api-auth, queue-process against a real WebUI.
2. **E2E (functional, real extension)** — load the actual built MV3 bundle, resolve the real extension id, drive the popup/options/content script end-to-end with user-observable assertions (torrent rendered, server saved+persisted, send→torrent appears in qBittorrent). The reference e2e is skeleton/non-functional (hardcoded test-id, no extension load fixture, missing global-setup).
3. **Automation / autonomous QA** — fully self-driving runs re-runnable N× (CONST §11.4.98 / §11.4.50), no manual intervention; HelixQA bank entries.
4. **Security / penetration** — credential-leak audits (CONST §11.4.10), XSS via injected magnet/dn into popup DOM, CSP of extension

pages, permission-scope minimization, MV3 host-permission abuse, malformed-magnet/.torrent injection, cookie/SID handling, encrypted-credential at-rest verification (the crypto module — for real, not a copy).

5. **DDoS / load** — many torrents detected on one page (1000s of magnet links), rapid scan cycles, burst of queue enqueues, API request flooding/rate-limit behavior under load.
6. **Scaling** — large .torrent files, large queue (1000s items + persistence), large detected-set rendering in popup, many configured servers.
7. **Chaos** — fault injection: chrome.storage failures/corruption mid-write, network drop/timeout/reorder during queue processing, server-worker termination mid-operation, corrupt persisted state recovery, partial-write faults (CONST §11.4.85 closed-set: process-death, network-fault, input-corruption, resource-exhaustion, state-corruption). The current chrome-mock is happy-path only — needs an error-injecting variant.
8. **Stress** — sustained load ( $N \geq 100$  iters /  $\geq 30$ s), concurrent contention ( $N \geq 10$  parallel enqueues/scans), boundary inputs (empty/max/off-by-one) with latency p50/p95/p99 captured (CONST §11.4.85).
9. **Performance** — only `crypto.test.ts` has perf asserts (and on a copy). NEEDED: parse-time benchmarks (bencode/torrent/magnet on large inputs), scan latency on heavy DOMs, popup render time, with captured evidence.
10. **Benchmark** — formal benchmark suite with baselines + regression detection (separate from inline perf asserts).
11. **UI tests** — beyond DOM-presence: actual rendering correctness, badge counts, list rendering of detected torrents, modal interactions producing state changes (current e2e asserts presence, not behavior).
12. **UX tests** — flows from the end-user's perspective: detect→select→send happy path, error feedback, empty/loading/error states, accessibility (a11y), keyboard nav, the configured `chrome.commands` shortcuts.
13. **Challenges** (`./challenges/scripts/`) — CONST §11.4.27(B), driving the real built extension against a real qBittorrent + Boba proxy, asserting user-observable outcomes (no false success).
14. **HelixQA banks + autonomous QA sessions** — CONST §11.4.27 — registered test banks per surface, autonomous session execution with captured evidence per check.
15. **Anti-bluff remediation across existing types** — every test must fail against a no-op stub and assert user-observable outcomes (CONST §11.4 / §11.4.1). The reference contains explicit bluffs (the

`expect(true).toBe(true)` Auth + auto-processing blocks; the crypto copy-not-module; non-functional e2e; dead-assert lines) that violate this — they are anti-patterns to fix, not patterns to copy.

---

## Open questions (for the plan)

1. **Runner:** standardize on **Vitest** (matches Boba frontend/ + `crypto.test.ts`) or **Jest** (matches the other 6 unit files)? Coverage config and ESM handling differ.
2. **Coverage threshold:** reference enforces **80%** and EXCLUDES all UI/entrypoint dirs. Boba mandates ~100% across types — confirm the line/branch threshold AND whether popup/options/content/background must be brought INTO coverage (they must, per §11.4.27).
3. **E2E extension-loading fixture:** need a Playwright fixture that builds `./dist`, launches a persistent Chromium context with `--load-extension`, captures the real extension id (via the service worker target), and parameterizes `chrome-extension://<id>/...`. Does Boba target Chromium only, or must Firefox MV3 (different load mechanism) be supported? The reference's Firefox project is broken for extensions.
4. **Live backends for integration/challenge:** which real services do extension tests run against — qBittorrent WebUI (:7185 via proxy :7186), merge service (:7187), webui-bridge (:7188), boba-jackett (:7189)? CONST §11.4.27 requires real services for all non-unit types.
5. **global-setup.ts/global-teardown.ts** referenced by playwright config but absent — define their contract (build extension, start backend containers, seed test data, capture extension id).
6. **Crypto module path:** reference comment says `src/utils/crypto.ts`; `queue/torrent` imports use `src/shared/*`, `src/parser/*`, `src/api/*`. Confirm the production crypto module path Boba will use, and rewrite crypto tests to import it.
7. **Chrome mock error-injection:** chaos/stress need a configurable mock that can throw / return failures / corrupt storage — extend `chrome-mock.ts` or build a separate fault-injecting harness?
8. **invalidMagnets fixture:** wire the 7 unused negative cases into magnet tests (each must throw or return null).
9. **Manifest version source:** `chrome-mock` hardcodes `name: "BobaLink", version: "1.0.0"` — does the plan rename to a Boba-branded extension, and should tests read the real manifest?

10. **CI:** Boba CI is manual (`./ci.sh`) and forbids `.github/workflows`. The playwright config's reporter: `github + forbidOnly: !!CI` assume GitHub Actions — must be reworked for Boba's manual-CI model.